

Sampling

1 Representation by Samples

Theorem 1.1 (Nyquist-Shannon Sampling Theorem).

Let $x(t)$ be a **band-limited** signal with a maximum frequency component of ω_M , i.e. $X(j\omega) = 0$ for all $|\omega| > \omega_M$. Then, $x(t)$ can be completely determined by its samples $x[n] = x(nT)$, where T is the sampling period, if the sampling frequency $\omega_s = 2\pi/T$ satisfies the condition:

$$\omega_s > 2\omega_M$$

where ω_M is then called the **Nyquist frequency**, and $2\omega_M$ is the **Nyquist rate**.

1.1 Impulse Train Sampling

Sampling Function: $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$ with **sampling period** T_s and **sampling frequency** $\omega_s = \frac{2\pi}{T_s}$.

$$\begin{aligned} x_p(t) &= x(t)p(t) \\ &= \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s) \end{aligned}$$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} P(j\omega) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)$$

$$\begin{aligned} x_p(t) = x(t)p(t) &\xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega) \\ &= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \end{aligned}$$

For a periodic signal $x(t)$ whose Fourier series is a_k ,

$$\begin{aligned} x(t) &\xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k \\ p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) &\xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} b_k = \frac{1}{T_s} \\ x_p(t) = x(t)p(t) &\xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} c_k = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} a_{k-n} \cdot \frac{T}{T_s} \end{aligned}$$

1.2

Suppose $x(t)$ is a **band-limited** signal with a maximum frequency component of ω_M .

$$\begin{aligned} x(t) &\xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(j\omega) \\ x_p(t) := \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s) &\xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X_p(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \quad \text{where } \omega_s = \frac{2\pi}{T_s} \\ x_d[n] := x(nT_s) &\xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X_d(e^{j\omega}) = X_p\left(j\frac{\omega}{T_s}\right) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X\left(j\frac{\omega - 2k\pi}{T_s}\right) \end{aligned}$$

2 Reconstruction via Interpolation

Impulse train sampling is not practical.

2.1 Zero-Order Hold (ZOH)

2.2 First-Order Hold (FOH)

3 Aliasing

4 DT processing of CT signals

5 Sampling of DT signals